LAICIs

Local AI Checker Integration System

Autonomous Multi-Modal Deepfake Forensics, PDRM Mule Account Interception & BNM RMiT Compliance Agent Powered by **Tencent WorkBuddy**.

3.2s

AUTONOMOUS TRIAGE SPEED

Reduced from 45 mins manual inspection across 5 websites.

7 Tools

WORKBUDDY MCP CATALOG

Parallel tool orchestration across vision, text & banking.

RM 1.22M

SIMULATED LOSS INTERCEPTED

Actuarially calibrated against PDRM CCID median benchmarks.

The High-Stakes Bottleneck: Fragmented Fraud Triage

❌ The Current Manual Nightmare (45 Mins)

- **Siloed Media Tools:** Uploading image to one site, checking metadata on another.
- **Blind to Threat Context:** Deepfake detectors don't understand extortion messages.
- **Zero Banking Integration:** No automated lookup against PDRM Semak Mule databases.
- **Compliance Lag:** Manual checking against BNM RMiT mandates misses the golden window.

✅ The LAICIs Solution (3.2 Seconds)

- **One Input, One Goal:** Paste text, URL, or upload media—the agent handles everything.
- **Autonomous Tool Selection:** Dynamically dispatches only relevant forensic tools.
- **PDRM CCID Integrated:** Automatically queries known mule accounts and syndicates.
- **Immediate Deliverables:** Downloadable PDRM police dossier & BNM audit report ready for law enforcement.

7-Stage Autonomous Directed Acyclic Graph (DAG)

True task decomposition, parallel execution, and automated self-validation with zero user micro-prompting.

01 Understand Input → 02 Autonomous Planning → 03 Parallel Tool Invocation → 04 Evidence Cross-Validation

05 Findings Synthesis → 06 Deliver Usable Report → 07 Financial Exposure & Prevention (RM)

Dynamic Error Handling

Automatic failover heuristics: if cloud vision APIs timeout, local PyTorch CNN & ELA models guarantee zero downtime.

Multi-Evidence Fusion

Cross-correlates ELA noise variance with PyTorch synthetic probability and Gemini multimodal visual inspection.

Calibrated Risk Bounds

Calculates financial exposure only when monetary intent is detected, completely eliminating false financial alarms.

End-to-End WorkBuddy Utilization: From Expert Requirements to MCP Runtime

🤖 Phase 1: WorkBuddy AI Experts (Specification)

We leveraged specialized Tencent WorkBuddy persona experts to architect the complete solution:

- **Product Architect Expert:** Synthesized comprehensive 2,600+ line PRD with user personas & acceptance criteria.
- **Business Analyst Expert:** Formulated BRD quantifying RM 1.3B fraud stakes & actuarial ROI.
- **Systems Engineering Expert:** Designed SRS defining the 7-stage DAG & MCP JSON-RPC schemas.

⚡ Phase 2: WorkBuddy MCP Server (Runtime)

Direct platform utilization via official protocols and enterprise authentication:

- **Tencent Cloud Gateway:** Authenticated with `SecretId` & `SecretKey`.
- **WorkBuddy MCP Protocol:** Model Context Protocol endpoint at `/api/workbuddy/mcp`.
- **7 Registered Tools:** PyTorch CNN, ELA, Sightengine, Gemini 3 Flash, PDRM Mule, BNM RAG, Financial Assessor.

Demo 1: Multimodal Deepfake Forensics

User Input

File: `synthetic_executive_headshot.png`
Goal: "Verify if this executive announcement is authentic."

`WorkBuddy Agent: Media Route Selected`

Autonomous Agent Findings

- **EfficientNetV2 CNN:** 100.0% synthetic feature probability.
- **ELA Anomaly Map:** High-frequency residual noise mismatch.
- **Gemini Multimodal:** Structural artifact along facial perimeter.
- **Financial Exposure:** RM 0.00 (Calibrated: Media-only, zero monetary solicitations).

```
[WorkBuddy Agent] Routing input_type: media...
[Plan] Dispatching [pytorch_efficientnetv2, local_ela_analyzer, sightengine, gemini_multimodal]
[PyTorch] Weights loaded (best_model.pth) → Synthetic probability: 1.000
[ELA Analyzer] Generated error-level heatmap → ELA anomaly confirmed
[Gemini Vision] Model: gemini-3-flash-preview → Boundary inconsistency flagged
[Consensus] Composite Score: 67.7% → Verdict: SUSPICIOUS_ANOMALIES
[Audit Logger] SHA-256: 4f9e1c... Logged under BNM RMiT compliance
[Status] HTTP 200 completed in 3.12s. Deliverables generated.
```

Demo 2: PDRM Mule Account & Extortion Triage

User Input (Text Message)

"URGENT: Your Maybank account 514012345678 has been frozen due to suspicious activity. Transfer RM 5,000 immediately to officer mule account 164012345678 to prevent arrest by PDRM."

🚨 Autonomous Triage Results

- **Verdict:** HIGH_RISK_SCAM (Risk Score: 90.0%)
- **PDRM Mule Hit:** Account 164012345678 flagged in CCID database.
- **Gemini Reasoner:** High urgency, police authority impersonation detected.
- **Exposure:** RM 1,128.00 | Immediate NSRC 997 hotline protocol initiated.

```
[WorkBuddy Agent] Routing input_type: text...
[Plan] Selecting [pdrm_ccid_semak_mule, gemini_scam_analyzer, financial_risk_assessor]
[PDRM CCID] MATCH FOUND: Account #164012345678 listed in CCID Mule Registry!
[Gemini LLM] Reasoning: Police impersonation + financial extortion tactics
[Financial Engine] Tier: MODERATE → Potential Exposure: RM 1,128.00
[RAG Synthesizer] Grounded citations: [BNM-RMIT-01], [BNM-FRAUD-02]
[Deliverable] Police Investigation Dossier prepared for PDRM filing.
```

Demo 3: Chrome & Edge Extension (Manifest V3)

🌐 Active Tab Inspection

Automatically reads current browser tab URL and queries domain reputation and fake portal databases in 1 click.

💬 Instant Message Scan

Paste WhatsApp texts or bank alerts directly into the popup for real-time Gemini & PDRM verification.

🖱️ Right-Click Context Menu

Right-click any web image → *"Scan Image for AI Deepfakes with LAICIs"* launches instant forensic report.

Live Backend Telemetry Integration

Automatically connects to `http://localhost:5000/api/agent/status` with live green status badge.

● ONLINE

Built-in Quality Control & Official Deliverables

🛡️ Automated Verification & Quality Control

- **Cross-Model Consensus:** Forensic scores fuse PyTorch CNN, ELA noise residuals, Sightengine, and Gemini visual inspection.
- **Non-Hallucinatory Financial Bounds:** Risk exposure strictly calculated via actuarial curves; RM 0.00 for non-monetary media.
- **Grounded Statutory Citations:** Regulatory RAG answers explicitly reference `[BNM-RMIT-01]` and `[BNM-FRAUD-02]`.
- **Cryptographic Audit Trail:** Every scan is stamped with an immutable SHA-256 hash in SQLite.

📄 Immediate Operational Deliverables

- **PDRM CCID Police Dossier:** Formatted for immediate submission to the Commercial Crime Investigation Department (`/api/reports/download/{id}?type=pdrm`).
- **BNM RMiT Audit Markdown:** Full regulatory report documenting MTI (Maximum Tolerable Downtime) and fraud kill switches.
- **NSRC 997 Rapid Protocol:** Step-by-step 24-hour golden window action guide.

Real-World Impact Across 3 Segments

👤 Individuals & Campus

Free Chrome extension to protect students and citizens from fake job offers, romance scams, and scholarship phishing.

Zero Friction · 1-Click Verification

🏢 Small & Medium Enterprises

Protects accounting teams against CEO impersonation, fake supplier bank account changes, and fraudulent invoices.

Prevents BEC Extortion

🏛️ Financial Institutions

REST API integration with automated compliance reporting adhering to Bank Negara Malaysia's RMiT standards.

Auditable · Cryptographic Trail

93%
REDUCTION IN TRIAGE TIME

100%
VERIFIED PDRM MULE MATCHES

0%
MICRO-PROMPTING REQUIRED

One Prompt. One Goal. Real System Actions Done.

LAICIs proves that Tencent WorkBuddy can solve Malaysia's greatest digital challenge—protecting citizens and businesses from synthetic fraud before money leaves the bank.

🚀 Live System

Fully operational backend daemon, React-free vanilla frontend, and Manifest V3 extension.

🤖 Autonomous Core

Tencent WorkBuddy MCP Protocol + Google Gemini + EfficientNetV2 PyTorch.

MY Sovereign Standards

Direct compliance with BNM RMiT, NSRC 997, and PDRM CCID frameworks.

DORMATHON 2026 · TEAM LAICIs · TRACK 2 (WORKBUDDY AGENTIC SOLUTIONS)